

Electron Microscopy

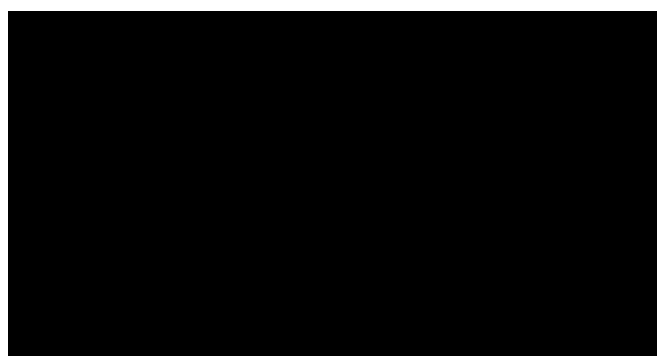
In contrast with light microscopy, electron microscopy uses electrons, instead of photons, to illuminate a sample. There are several advantages to this approach: Because electrons have much shorter wavelengths than photons, they yield a much higher resolution, allowing much greater magnification than is possible using light—around 1,000 times greater magnification! Depth of field is also not an issue, even at high magnifications, allowing excellent resolution of three-dimensional objects. The limitations of electronic microscopy are also significant: Electrons require a vacuum to operate; samples must be capable of conducting an electric charge, so they must be coated with materials such as gold or graphite; electron microscopes cost at least tens of thousands of dollars. A new generation of “tabletop” electron microscopes has significantly brought down the cost of such devices, but they remain very expensive items that only specialized laboratories can afford.

Just as light microscopy can be divided between transmitted and reflected light, so too can electron microscopy. Transmission electron microscopy, often abbreviated TEM, is used in some forms of material analysis but rarely is applied to understand archaeological materials, as it requires that materials studied be incredibly thin—one ten-thousandth of a millimeter or less! In contrast, scanning electron microscopy, or SEM, can be used on many different kinds of materials and has found a number of uses in archaeology, including in the study of archaeological evidence for foodways.

Scanning Electron Microscopy

A scanning electron microscope uses a concentrated beam of electrons that rapidly sweeps the surface of a three-dimensional object. If the object is conductive, it will generate low-energy **secondary electrons**, which are recorded with a specialized detector in the microscope and converted to a digital signal that displays on a monitor, giving an image of the surface.

Watch this short video to see an SEM in action!





Secondary electrons are produced when the scanning electron beam sweeps over a conductive surface. They provide topographical information about the surface of the sample being analyzed.

SEM-EDS

SEM also produces higher-energy interactions deeper in the material of the sample; such interactions produce X-rays from the atoms of the sample. The wavelength of these X-rays is characteristic of the type of atom being illuminated. An X-ray detector within the microscope can distinguish between these X-rays and thus assess the elemental composition of the sample, giving information on what the sample is made of on a point-by-point basis. The relative intensity of X-rays characteristic of different elements can be compared, allowing inferences about the relative concentration of the elements in the sample. This method is known as energy-dispersive X-ray analysis, or EDS (sometimes EDX). The combination of SEM and EDS is known as SEM-EDS.

If you are interested in learning more about the atomic physics involved, see this short, illustrated web page from [ThermoFisher](https://www.thermofisher.com/us/en/home/global/forms/industrial/edx-analysis-sem.html) (<https://www.thermofisher.com/us/en/home/global/forms/industrial/edx-analysis-sem.html>), a manufacturer of SEM-EDS systems.